Adventist University of Central Africa

BUILDWISE PROJECT CONTROL SUITE

Case study: Adroit Construction Company Ltd

A final year project presented in partial fulfilment of the requirements for the degree of BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

Major in Software Engineering

By

Nshuti Aimable

February, 2026

ABSTRACT

Adventist University of Central Africa

A final-year project Presented in partial fulfilment of the Requirements for the degree of bachelor of science in information technology

Major in Software Engineering

Title: BUILDWISE PROJECT CONTROL SUITE

Name of the student: Nshuti Aimable

Date Completed: February, 2026

The management of construction projects is a critical driver of economic growth, but firms in Rwanda face significant challenges with efficiency and oversight. Traditionally, project data including schedules, budgets, and material procurement has been managed through fragmented manual processes and decentralized spreadsheets, resulting in frequent delays and cost overruns. Although enterprise-grade digital solutions exist globally, they often present a "complexity gap" for regional contractors due to high costs and workflows that are not optimized for the local market. This project was created to close that gap by developing the BuildWise Project Control Suite, a platform designed to offer centralized, automated, and efficient project monitoring for Adroit Construction Company Ltd.

The challenge was replacing reactive, uncoordinated processes with seamless site-to-office coordination. The platform achieves this through centralized Gantt charts, automated procurement, budget workflows, and real-time progress dashboards.

Data was gathered via stakeholder workshops to map operational workflows. Developed iteratively using Agile two-week sprints, the system features a user-centered ReactJS and Tailwind CSS frontend, powered by a secure Python and PostgreSQL backend.

This unified web-based suite enables real-time collaboration between project managers, site engineers, and procurement officers. It enhances transparency and operational efficiency, ensuring construction projects are delivered with data-driven precision.

DECLARATION

I, **Nshuti Aimable** with Registration Number 24854, a student at the Adventist University of Central Africa, in the Faculty of Information Technology, Department of Software Engineering.

I hereby declare that this project, titled as "**BuildWise Project Control Suite**" is my own original work. I have carried out this project through personal research, development, and effort, and it has not been submitted for academic credit at AUCA or any other institution.

Signature: …….

Date: …. /…./….

DEDICATION

I sincerely thank God for His grace and strength throughout this journey.

My gratitude goes to my family for their unwavering encouragement.

To my supervisor and IT faculty lecturers for their guidance.

I also appreciate mental health advocates and professionals whose commitment continues to inspire positive change.

APPROVAL

I, Mr. MANIRAGUHA Eric, confirm that I have given my consent for this final year project report to be completed and submitted under my supervision.

Signature: …….

Date: …. /…./….

ACKNOWLEDGEMENTS

First and foremost, I thank God Almighty for His endless grace, wisdom, and strength that have guided me through this academic journey. Without His love and guidance, completing this project would not have been possible.

I am truly grateful to the administration and academic staff of AUCA for their dedication, patience, and commitment to nurturing our growth, as your teaching and encouragement have shaped both my learning and my character. My heartfelt thanks also go to the Faculty of Information Technology, especially the Department of Software Engineering, for providing the foundation, knowledge, and support that made this journey meaningful and rewarding.

I owe special gratitude to my supervisor, Mr. Eric Maniraguha, for his steady guidance, insightful feedback, and encouragement. Your mentorship not only helped me complete this work successfully but also inspired me to think critically and grow as a student and person.

I am profoundly thankful to my family, whose love, sacrifices, and support have been my greatest strength throughout my academic journey. Thank you for your patience during late nights, your wise guidance when I felt lost, and your belief in me even when I doubted myself. Your encouragement has been my motivation to keep going, and your values have shaped me.

To my siblings, thank you for your inspiration and continuous encouragement that pushed me to stay focused and achieve my goals.

I'm also deeply thankful to my colleagues for their support, laughter, and motivation. Sharing this journey with you made it lighter and much more enjoyable.

Finally, to everyone who supported me in one way or another thank you. This achievement is as much yours as it is mine.

May God bless you abundantly!

Nshuti Aimable

CHAPTER 1

GENERAL INTRODUCTION

Introduction

The construction industry is a cornerstone of national development, driving economic growth through the creation of essential infrastructure, housing, and industrial facilities. Managing these complex projects requires a delicate balance of time, labor, materials, and capital. In Rwanda, the construction sector is a primary driver of the national economy, yet many private firms continue to face significant structural challenges in project oversight. Sadly, project data is often managed through fragmented manual processes, physical logbooks, and decentralized spreadsheets, resulting in frequent delays, material waste, and increased operational costs.

With the global industry shifting toward Construction 4.0, new and innovative digital solutions have emerged to bridge this gap. Integrated Project Control Suites have shown great potential in offering centralized, real-time oversight and mitigating persistent challenges like budget overruns and communication silos. Unlike traditional paper-based workflows, these digital platforms can automate scheduling, provide instant financial variance analysis, and serve as a "single source of truth" for all stakeholders. They provide a reliable environment for project managers who need to anticipate risks before they escalate into costly delays.

This project, titled “BuildWise Project Control Suite,” was created to address these operational bottlenecks by developing a centralized digital management platform for Adroit Construction Company Ltd. The system aims to offer users interactive Gantt chart scheduling, real-time budget and cost control, automated material procurement, and a progress monitoring dashboard. Beyond basic data entry, BuildWise is designed to seamlessly connect the construction site with the head office, replacing reactive problem-solving with proactive, data-driven coordination.

The goal of this project is not just to build a technical solution, but to make construction project management more transparent, accountable, and operationally efficient. By digitizing the entire project lifecycle, BuildWise strives to empower stakeholders with the precise insights needed to prevent critical delays and resource wastage. This initiative also aligns closely with Rwanda’s national push toward digital excellence in all professional sectors and the broader

transition to global construction standards. Through this project, technology becomes more than an administrative tool it becomes a catalyst for precision, operational excellence, and sustainable infrastructure development.

Background of the Study

Construction is a cornerstone of national development, driving economic growth through the creation of essential infrastructure, housing, and industrial facilities. In Rwanda, the construction sector has become a primary driver of the national economy, reflecting the country's rapid development and broader push toward digital excellence in all professional sectors. Recognizing the importance of robust infrastructure, national digital initiatives have set a precedent for the efficiency of digital governance, laying the foundation for modernizing professional industries. This early commitment to modernization set the stage for a long-term effort to elevate the quality and efficiency of infrastructure development nationwide.

Globally, the industry is embracing "Construction 4.0" by adopting integrated digital suites to mitigate budget overruns, delays, and communication silos. However, digital adoption in Rwanda's private construction sector remains low. Many local contractors still rely on manual tools and ad hoc risk management because expensive, complex global platforms like Procore or Oracle fail to align with the procurement realities of the East African market.

Adroit Construction Company Ltd stands as a prominent construction and engineering firm located in Kigali, Rwanda. Since its establishment, the company has been responsible for delivering a wide range of infrastructure projects, including residential complexes, commercial buildings, and civil engineering works. However, the company operates under significant operational constraints. To sustain its operations, Adroit relies on a complex network of internal processes to manage labor, procurement, and heavy machinery. Project management activities are largely reactive, relying heavily on manual logbooks, physical document folders, and fragmented spreadsheets rather than integrated digital analysis. This creates a significant "information lag," leaving management with outdated data and preventing the timely detection of budget overruns or schedule slips.

In this context, digital, scalable solutions such as the BuildWise Project Control Suite are particularly relevant. Such a tool can address the urgent need for technological solutions that integrate project activities into a seamless digital workflow. By connecting the construction site directly to the head office, the suite can help reduce the administrative burden on the institution, improve real-time visibility, and deliver a cost-effective solution tailored for Rwandan firms.

This project responds directly to the need for innovation in Rwanda’s construction sector by proposing the development of a comprehensive digital management platform designed to automate the entire project lifecycle. By offering real-time financial monitoring, interactive Gantt chart scheduling, and automated material procurement workflows, the system helps bridge the temporal and geographic gaps between site engineers, procurement officers, and project managers.

Leveraging modern web technologies such as ReactJS, Python, and PostgreSQL, the system provides a dynamic, component-based architecture tailored to the operational realities of the local market. This addresses the mismatch observed in both generic task managers (which lack industry-specific "DNA") and highly complex enterprise ERPs that fail to resonate with regional contractors.

The solution is designed to complement existing operational frameworks, transforming reactive, paper-based workflows into a proactive, analytics-driven system. By introducing an integrated platform that works alongside human professionals—from on-site supervisors to executives—it helps strengthen Adroit Construction's internal operations and ensures that project inputs are synchronized with construction schedules.

Ultimately, the project aims to contribute to Rwanda’s national goal of enhancing infrastructure quality through the leverage of modern technology. It seeks to enhance transparency, strengthen accountability, and reduce the risk of project failure, advancing the company's capacity to deliver construction projects with higher precision and operational excellence.

Statement of the Problem

The aim of this study is to design and develop the BuildWise Project Control Suite, a comprehensive digital management platform that will modernize Adroit Construction Company Ltd's approach to project oversight by addressing gaps in scheduling, cost tracking, and operational coordination. The system will provide a centralized, web-based and desktop-compatible platform offering continuous monitoring, interactive Gantt chart scheduling, real-time financial tracking, and automated material procurement. By bridging the gap between site execution and office administration, this platform will directly address systemic inefficiencies, improving transparency, accountability, and the overall reliability of project data.

Key operational challenges facing Adroit Construction Company Ltd and its current project control system:

The current project control model at Adroit Construction faces several structural, financial, and technological challenges that limit effective service delivery and real-time decision-making.

- **Reliance on fragmented, manual processes:** One of the most pressing issues is the heavy dependence on paper-based logs and disconnected spreadsheets. Project schedules are often tracked through static documents, while financial expenditures and daily progress are recorded manually by site engineers. This workflow creates a significant "information lag," as data must be manually compiled and transported to the office weekly, leaving management with outdated information.
- **Inefficient project coordination and communication:** Limited accessibility and continuity of information further exacerbate operational problems. Site engineers often work from outdated task lists and lack immediate access to updated resource allocation plans. This uncoordinated communication between site teams, procurement officers, and managers leads to inefficient task execution, labor idleness, and duplicated effort.
- **Inaccurate cost tracking and economic waste:** Poor financial control remains a deeply rooted issue that significantly affects project profitability. Without real-time cost tracking, management often identifies budget overruns only after funds have been spent, preventing the implementation of timely cost-saving measures. Furthermore, the procurement process remains siloed, frequently leading to emergency material purchases at inflated costs or unnecessary overstocking.

- **Highly reactive problem-solving:** Another critical challenge is the inability to anticipate risks and intervene promptly. Due to delayed data collection, management cannot identify schedule slips or task delays in real-time. The company lacks the predictive capability needed to prevent bottlenecks, often addressing issues days after they have already slowed down the project's critical path.
- **Lack of relevant digital alternatives:** The effectiveness of existing digital construction tools is limited by a lack of operational relevance for regional contractors. Enterprise-grade solutions (like Procore or Oracle) present a "complexity gap" with high licensing costs and workflows unsuited for the East African market. Conversely, generic task managers lack the industry-specific "DNA" required for specialized construction tracking, such as Bill of Quantities (BOQ) and financial variance analysis.
- **Limited accountability and absence of a "source of truth":** Finally, there are weak oversight mechanisms to monitor daily site activities. Project managers must manually compile progress reports from various departments, a process that is highly prone to human error. The absence of a centralized digital repository and audit trails makes it difficult to track specific engineer accountability or monitor who authorized specific site expenditures.

The BuildWise platform addresses these challenges by centralizing project scheduling, financial monitoring, and procurement into a unified digital ecosystem. By replacing reactive, paper-based workflows with proactive, analytics-driven dashboards, the system improves accessibility to project data, reduces the risk of project failure, and strengthens the company's operational capacity. Ultimately, the platform aims to create a secure, highly efficient project control environment that aligns Adroit Construction with global Construction 4.0 standards and Rwanda’s push toward digital excellence.

Choice and Motivation of the Study

General Motivation: This project was inspired by the critical need to modernize construction management in Rwanda. Currently, many firms struggle with fragmented communication and inaccurate cost tracking due to outdated manual processes. BuildWise addresses these inefficiencies by providing a centralized, automated platform that ensures real-time financial tracking, interactive scheduling, and seamless site-to-office coordination.

To AUCA: This project reflects the academic rigor and values of the Adventist University of Central Africa (AUCA), which emphasizes the practical application of technology to drive national development and professional excellence. It allowed me to apply the technical knowledge gained throughout my studies in Software Engineering to solve a real-world industrial challenge. By integrating modern web technologies, complex database management, and user-centered design, this project demonstrates how AUCA students can contribute to the digital transformation of Rwanda's infrastructure sector, making construction management more transparent, efficient, and data-driven.

To Adroit Construction Company Ltd: Adroit Construction Company Ltd, a prominent engineering firm in Kigali, faces the growing challenge of managing complex projects with fragmented, manual oversight tools. The BuildWise platform acts as a comprehensive digital assistant, helping project managers and site engineers automate task assignments, monitor real-time budgets, and optimize material procurement. It also includes a Progress Monitoring Dashboard and automated alert features, which will significantly ease the administrative burden on the company’s staff and support proactive, data-driven decision-making across all active construction sites.

To the Student: For me, as a final-year Software Engineering student, this project has been both a significant technical challenge and a professional milestone, reflecting my belief that software should drive operational excellence and empower industries. It strengthened my skills in full-stack web development, relational database design, and Agile methodologies, and taught me the importance of creating robust systems that solve complex, real-world logistical problems. This work aligns directly with Rwanda’s push toward Digital Twins and Construction 4.0, representing my effort to inspire future AUCA innovators to build scalable solutions that make a meaningful economic impact.

Objectives of the Study

General Objective

The overall goal of this project is to design and develop an integrated, web-based and desktop-compatible project control system that automates the planning, monitoring, and financial tracking of construction activities at Adroit Construction Company Ltd. By leveraging innovative technology, the system seeks to address the gaps in the current manual project management landscape and provide a scalable, centralized digital suite that improves operational performance, minimizes administrative delays, and ensures cost-effectiveness through real-time collaboration between project managers, site engineers, and procurement officers.

Specific Objectives

- Digitize and streamline the entire project planning and scheduling process through a secure, centralized platform that incorporates interactive Gantt charts and automated milestone tracking. This will eliminate the traditional reliance on static spreadsheets, significantly reduce resource scheduling conflicts, and ensure that project timelines remain accurate, transparent, and easily adjustable to reflect dynamic, real-world site conditions
- Implement a comprehensive budget and cost control module that allows for real-time tracking of expenditures and resource utilization, providing a live comparison between actual and planned costs to enable the early detection of financial variances.
- Create a centralized task management and material procurement interface that facilitates seamless coordination between the site and the office, allowing site engineers to update task progress digitally and enabling procurement officers to manage material requests.
- Offer real-time analytics and interactive dashboards that support data-driven decision-making for senior management by visualizing performance metrics and project health indicators, allowing stakeholders to identify delays instantly.

- Establish a secure document management and communication system that ensures the integrity of blueprints, permits, and project files by implementing role-based access controls and automated notification services (SMS/Email) for critical alerts.

Scope of the Project

This study is limited by specific technical and operational boundaries that shape the functionality and applicability of the BuildWise Project Control Suite. The system includes the design and development of a web-based and desktop-compatible platform featuring ten core modules, such as Project Planning with Gantt charts, Budget & Cost Control, and Material Procurement. While the system offers robust digital oversight, real-time collaboration, and interactive data visualization for Adroit Construction Company Ltd, it remains strictly confined to core operational monitoring and control. The platform does not incorporate advanced artificial intelligence features, such as automated predictive risk modeling or machine-learning-based cost estimation.

The functionality of the suite also depends heavily on stable internet connectivity, as real-time collaboration and cloud-syncing features require network access at construction sites. Additionally, the project scope strictly excludes the procurement or installation of physical hardware, such as on-site servers, networking equipment, or specialized IoT sensors for tracking heavy machinery. The system also does not include full integration with external third-party software, such as national tax authority systems or proprietary enterprise accounting suites like SAP or Oracle, operating instead as a standalone project control environment.

In terms of operational rollout, large-scale training for the entire company workforce falls outside the initial scope; implementation will be limited to "Train-the-Trainer" sessions for department heads and system administrators. Furthermore, the system's output is highly dependent on the timely and accurate entry of data from the field; manual entry errors can still impact the integrity of reporting and analytics. Long-term hosting fees, server maintenance, and operational support beyond the agreed-upon pilot phase are also excluded from this study.

Overall, these boundaries highlight the system as a focused, operational digital tool rather than a fully automated, hardware-integrated enterprise resource planner. Future system improvements may address these gaps by expanding functionality to include offline-capable

features for data entry in remote locations, hardware sensor integrations, and advanced predictive analytics.

Methodology and Techniques Used in the Study

This project uses a mixed-methods approach, combining qualitative research techniques with the Agile software development methodology. This ensures a well-rounded process for designing, developing, and validating the system, emphasizing iterative progress and continuous feedback.

Documentation

Documentation is a qualitative research method that involves systematically collecting, reviewing, and analyzing existing written materials to understand the background of a problem and identify existing challenges. In this study, I used documentation to review the existing fragmented manual workflows at Adroit Construction Company Ltd, specifically focusing on physical logbooks and disconnected spreadsheets. This review provided important insights into the "information lag" created when data is manually compiled and transported to the office, highlighting the vulnerabilities of reactive problem-solving.

Interviews and Stakeholder Workshops

In this study, structured stakeholder workshops were conducted with Adroit Construction's project managers and site engineers. Officially approved by company leadership , these sessions aimed to map site-to-office workflows and identify real operational bottlenecks.

I, Nshuti Aimable, a final-year student at AUCA in the IT Faculty, Major in Software Engineering, am conducting this project on the development of the BuildWise Project Control Suite. The project seeks to improve operational performance, minimize administrative delays, and ensure cost-effectiveness through digital innovation.

Below are the key themes and summarized responses gathered during the operational analysis and workshops:

- **Question 1: What are the main challenges faced during daily project control?**
    - *Answer:* Participants highlighted that the reliance on paper-based logs and disconnected spreadsheets creates a fragmented workflow. The manual transfer of data from the site to the office results in outdated information and delayed decision-making.
- **Question 2: How does the lack of a centralized system affect project finances?**
    - *Answer:* Respondents explained that without real-time tracking, budget overruns are often identified only after the funds have been spent. Additionally, uncoordinated material procurement frequently leads to overstocking or emergency purchases at inflated costs.
- **Question 3: What role do you think a digital platform could play in improving site operations?**
    - *Answer:* Participants felt that a centralized digital solution would replace paper-based workflows, ensuring greater efficiency and financial accuracy. It would allow site engineers to provide immediate progress updates through a user-friendly interface.
- **Question 4: What features are most critical for the proposed system?**
    - *Answer:* Suggested features included interactive Gantt charts for scheduling, real-time financial monitoring with actual vs. planned cost comparisons, and centralized material procurement workflows.
- **Question 5: What concerns exist regarding the implementation of a digital suite on construction sites?**
    - *Answer:* Participants expressed concerns about internet dependency, as real-time syncing requires stable connectivity at remote sites. They also highlighted the need for training, as some site-level personnel might struggle to transition from traditional paper logs to a digital interface.

Expected Results

Expected results refer to the anticipated business impacts and measurable improvements that the implementation of the BuildWise Project Control Suite will bring to Adroit Construction Company Ltd.

Upon successful implementation of the system, the following outcomes are expected:

- **Efficiency Gains:** Automated scheduling and real-time task updates will reduce administrative overhead. This will accelerate project delivery by minimizing information silos.
- **Enhanced Financial Control:** Real-time cost-variance analytics will enable the early detection of budget overruns. This allows management to take immediate corrective actions.
- **Improved Transparency:** Centralized material and labor tracking will provide stakeholders with a clear audit trail. This ensures all project expenditures and site activities are visible and accountable.
- **Data-Driven Decision Making:** Interactive dashboards will provide leadership with actionable insights. These insights will cover project health, resource utilization, and potential risks.
- **Comprehensive Documentation:** The system will automatically generate technical reports. It will also maintain version-controlled archives of all project files, ensuring long-term institutional knowledge.
- **Relief of Administrative Burden:** Project managers will be relieved from the burden of manual data reconciliation. Site engineers will be able to easily provide progress updates through optimized interfaces.

Organization of the work

- **Chapter One:** Introduces the study by providing background on construction management challenges, the problem statement, motivation, objectives, scope, methodology, and anticipated outcomes.
- **Chapter Two:** Examines the current project control landscape, reviews literature on Construction 4.0, and compares existing global solutions (like Procore) against the localized needs of Rwandan contractors.
- **Chapter Three:** Focuses on the system design, detailing the AS-IS manual process model, the proposed TO-BE digital workflow, and the overarching system architecture and database schema.
- **Chapter Four:** Covers the Agile development and testing of the BuildWise platform, detailing the use of ReactJS, Python, and PostgreSQL, alongside the integration of core modules like Gantt charts and budget tracking.

- **Chapter Five:** Summarizes the expected outcomes, reflects on the system's potential limitations (such as internet dependency), and highlights recommendations for future scalable deployment across multiple sites.